CALIFORNIA
Water Boards

Edmund G. Brown Jr.
Governor

Matthew Rodriquez
Secretary for
Environmental Protection

North Coast Regional Water Quality Control Board

June 7, 2018

Attn: Steven Roper
College of the Redwoods
7351 Tompkins Hill Road
Eureka, CA 95501
Steven-Roper@Redwoods.edu

Dear Mr. Roper:

Subject: College of the Redwoods Wastewater Treatment and Disposal System, Eureka, Humboldt County
Notice of Applicability of Water Quality Order WQ 2014-0153-DWQ

File: College of the Redwoods, 7351 Tompkins Hill Road, Eureka, CA
WDID No. 1B80121OHUM; CIWQS Place ID No. 215141

The College of the Redwoods (College) submitted a report of waste discharge (ROWD) dated January 31, 2015, for the College of the Redwoods wastewater treatment facility in Humboldt County. The College provided supplemental information in April 2016. Based on the information presented in the ROWD and supplemental information, the wastewater treatment and disposal system and discharge are consistent with the requirements of the State Water Resources Control Board (State Water Board) General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems, Order WQ 2014-0153-DWQ (General Order).

This letter serves as Notice of Applicability (NOA) from the North Coast Regional Water Quality Control Board (Regional Water Board) that the General Order applies to the subject wastewater and disposal system and that the system, upon commencement of operation, is henceforth regulated under the General Order as described below. A Monitoring and Reporting Program (MRP) for the wastewater system is included as an attachment of this NOA. A revised NOA will be issued once the new facility has been constructed and upon receipt of an up to date Report of Waste Discharge.

The College will be upgrading their treatment facility and constructing a new disposal area, scheduled for completion in 2020. See “Facility and Discharge Description” below for a description of the new facility. The College requested this NOA to procure grant funding and proceed with construction of the new facility. Table 1 provides a tentative schedule for milestones for constructing the new facility.

Table 1. Anticipated Milestone and Completion Dates for College of the Redwoods Wastewater Treatment and Disposal System

Milestone	Planned Completion By	Current Status[1]
Final Chancellor's Office Approval	02/2018	Completed
Bidding	04/2018	Occurring
Contracting	08/2018	Not Started
Construction	12/2019	Not Started
Startup and Testing	02/2020	Not Started
Closeout	06/2020	Not Started
[1] As of May 2018		

The General Order contains requirements that apply to all wastewater systems and requirements that apply only to specific types of treatment and/or disposal systems. Applicable requirements and effluent limits are discussed in Section 3 of this NOA and are summarized in Table 2 in Appendix A. The General Order contains additional operational and reporting requirements that are not summarized in the table. The Discharger is responsible for all the applicable requirements in the General Order, this NOA, and the MRP.

1. FACILITY AND DISCHARGE DESCRIPTION

Facility
The College of the Redwoods (hereafter "Discharger") owns and operates a wastewater treatment facility (Facility) located in Eureka, Humboldt County. Because the College is not located in an area easily served by a regional wastewater collection system, the College treats and disposes of wastewater generated on campus on-site. Attachment A to this NOA shows the site location.

The College is located on a 156.4-acre parcel (APN No. 307-01-114) and consists of residence halls, campus dining facilities, multiple science labs, maintenance buildings, a community stadium, a greenhouse and additional infrastructure. The Facility is currently regulated under National Pollutant Discharge Elimination System (NPDES) Permit No. CA0006700 (Waste Discharge Requirements Order No. R1-2016-0041) and Cease and Desist Order No. R1-2016-0043. The Facility has a permitted design flow of 0.08 million gallons per day (MGD). Attachment B to this NOA provides a site plan.

The volume of waste entering the Facility varies widely depending on the time of year and weather conditions. During the school term (generally, August through May), the campus may have a daily occupancy of up to 6,000 people who use facilities during normal class times. This can increase by several hundred during special events. The campus daily occupancy drops under 100 persons during certain holiday weekends and academic breaks.

Attachment C to this NOA provides a schematic of the Facility. The Facility is composed of an influent septic tank, a recirculating tank, two recirculating gravel filters, two leach fields, and support facilities.

Discharge Description

The Facility is designed for a peak flow of 0.08 MGD. Peak daily flow conditions are expected to take place when class is in session and the maximum number of students and faculty are present, and during special events.

The *Utility Replacement Seismic Strengthening Project Design Report, for College of the Redwoods*, prepared by GHD Engineering, and dated June 2015 (Report), provides a plan layout of the wastewater collection system. This Report is made part of this NOA by reference.

Collection System and Primary Treatment

The majority of the campus wastewater is collected and delivered to the wastewater treatment plant via a standard gravity collection system. The collection system has one pump station that serves the Child Development Center and a Snack Shack. The proposed system upgrade will include replacement of this pump station.

Primary sewage treatment is provided by an 80,000-gallon three-chambered cast-in-place septic tank. The influent septic tank allows for pretreatment through liquids/solids separation which significantly reduces biochemical oxygen demand (BOD) and total suspended solids (TSS) loading on downstream treatment units. In addition, the septic tank provides an anoxic environment for organic nitrogen conversion and denitrification. The design engineers indicate that the septic tank is specifically designed to handle the peak daily design flow of 80,000 gallons. Therefore, the minimum hydraulic detention time for the design flow event will be one day. Given its size, the septic tank is expected to afford a degree of attenuation of diurnal peaks as well as wet weather peaks. The solids in the septic tank will be periodically pumped and hauled to an appropriate offsite location for treatment and disposal.

The College will monitor influent flow following the septic tank, using a v-notch weir and ultrasonic level system to measure the range of flows. This configuration is intended to reduce the problems associated with fouling of flow measuring systems with raw wastewater influent solids and debris that are problematic in traditional flumes during low flow periods. Due to the variable nature of the population on the campus throughout the day and by season, very low flows are expected to occur diurnally as well as during breaks in the academic year. Although the septic tank will attenuate flows to some degree, the septic tank effluent flow is expected to be representative of the wastewater flow characteristics.

Secondary Wastewater Treatment

After flow measurement, septic tank effluent is conveyed to a 40,000-gallon recirculation tank. The recirculation tank provides additional storage in the treatment system to help attenuate peak flows. Additionally, the tank provides an anoxic environment for continued denitrification.

After the recirculation tank, the effluent goes to two recirculating gravel filters in parallel, which provide the main secondary treatment for the Facility. The recirculating gravel filters are dosed with wastewater from the recirculation tank using two recirculation pumps (one on duty and one on standby). The filters provide an aerobic environment which promotes nitrification as well as further BOD and TSS removal through various physical and biological mechanisms. The recirculating gravel filter system is composed of two gravel filters, with a total filter surface of approximately 13,200 square feet (sf). The design engineer reports that the hydraulic loading rate used will provide overall robustness to the design and afford a factor of safety for handling periodic peak flows that may occur. The yard piping is configured to return flow from the gravel filter underdrain to the septic tank to aid in the denitrification process prior to disposal to the leach field.

Wastewater Discharge System

The recirculation tank is equipped with a recirculating splitter valve. As the level of water in the recirculation tank rises, it will cause a floating ball valve to shut off the water returning from the recirculating gravel filters and will divert it to the effluent pump station wet well. The wet well is equipped with two pumps (one in use and one on standby). Rising water level in the effluent pump station wet well will start the lead effluent pump and pump down the wet well to the low level shut off. In the event of unusually high flows, the standby pump can run as well. Effluent pumped to the leach field will not be disinfected. The effluent pumps will discharge to a 6-inch diameter force main that is routed approximately 3,300 feet along the Facility's property line and the Humboldt County Right of Way to the leach field location.

The leach field site is located behind the existing stadium, approximately 70 feet uphill from the treatment facility. The design engineer selected this site following a campus wide search for suitable leach field locations that would have the right combination of available area, compatible uses, soil conditions, groundwater characteristics, percolation characteristics, and other factors.

There are two leach fields for effluent distribution. Field 1 is 921 feet long and Field 2 is 931 feet long. Each trench has 11 feet of sidewall below the leach line. This results in 21,183 square feet of infiltrative area in Field 1 and 21,413 square feet of infiltrative area in field 2. Indexing distribution valves are used to alternate application between leach lines with each cycle of effluent pumps. This arrangement allows soil moisture levels to recover prior to receiving more effluent.

General WDR Order Coverage

From the date that the new facility comes online and begins discharging to the leach field, scheduled for June 2020, waste discharges from the Facility are regulated under the General Order (Order WQ 2014-0153-DWQ).

2. CALIFORNIA ENVIRONMENTAL QUALITY ACT

As an existing facility, the facility upgrade project described in this NOA is categorically exempt from the California Environmental Quality Act (CEQA) pursuant to California Code of Regulations, title 14, section 15301, (ongoing or existing projects), section 15302 (replacement or reconstruction of existing utility systems), and section 15303 (new construction or conversion of small structures).

3. SITE-SPECIFIC REQUIREMENTS AND EFFLUENT LIMITS

A. General

The General Order contains prohibitions and specifications that apply to all wastewater systems, and specifications that apply only to specific types of treatment and/or disposal systems. Table 2 of this NOA provides a summary of applicable discharge requirements for the Facility and associated effluent limits.

Facility managers and operators must be familiar with the applicable requirements contained in the General Order, this NOA, and MRP R1-2018-0039.

A copy of the General Order is enclosed with this NOA. The General Order is also available on the Internet at:

https://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

The wastewater treatment operator must be familiar with the requirements contained in the General Order, this NOA, and the MRP.

B. Reports to be submitted to the Regional Water Board

The following technical reports, which form the main components of a comprehensive Operation, Maintenance, and Monitoring program, shall be submitted for the Executive Officer' approval as described below:

1. By December 31, 2020, the College shall submit a Spill Prevention and Emergency Response Plan (Response Plan) consistent with the requirements of General Order Provision E.1.a. The Response Plan shall, at a minimum, address the following:

   a) Operation and Control of Wastewater Treatment - A description of the Facility's equipment, operational controls, flow measurement and calibration procedures, and treatment system schematic including valve/gate locations.

   b) Sludge Handling - A description of the sludge handling equipment, operational controls, and disposal procedures.

   c) Collection System Maintenance – A description of collection system cleaning and maintenance, equipment tests, and alarm functionality tests to minimize the potential for wastewater spills originating in the collection system or headworks.

   d) Emergency Response - A description of emergency response procedures including for emergencies such as power outage, severe weather, flooding, or inadequate freeboard (for systems with wastewater or recycled water ponds). An equipment and telephone list for contractors/consultants, emergency personnel, and equipment vendors.

   e) Notification Procedures - Coordination procedures with fire, police, Governor's Office of Emergency Services (CalOES), Regional Water Board, and local county health department personnel.

2. By December 31, 2020, the College shall submit a Sampling and Analysis Plan consistent with the requirements of General Order Provision E.1.b.

C. Septic Tanks

The Facility includes a septic tank that provides primary treatment of the wastewater. The Septic tank requires proper operation, maintenance, and monitoring in order to provide adequate and reliable wastewater service. This includes periodic inspection and removal, as needed, of accumulated solids. General Order Section B.2, Septic Systems, and the enclosed MRP, at section III, Septic Tank Monitoring and Maintenance include requirements for the septic tank. Qualified service providers may conduct operational maintenance and monitoring on the septic system with appropriate oversight by the College. The College is responsible for ensuring proper operation, maintenance, and monitoring of the septic tank.

D. Subsurface Disposal System Requirements

The Facility's wastewater treatment/disposal process includes dispersal of treated effluent to subsurface soil via an underground disposal area. The disposal area is made up of 920 feet of total trench length distributed over nine parallel percolation lines 20 feet on center for the primary field. The College shall ensure that it maintains a constructed primary dispersal system with capacity for 100 percent of design flows, and a 100 percent capacity designated reserve area. The reserve area is for future use, in case the existing disposal system is found to be inadequate. The reserve area shall remain available until needed. The College has designated a 100 percent capacity reserve disposal area adjacent to the primary disposal area.

E. Effluent Limits

Based on the septic tank and recirculating filter (trickling filter) system employed, the discharge from the Facility is subject to technology-based effluent limits. Table 2 presents BOD numeric limits. For trickling filter wastewater systems, the total suspended solids limit is not appropriate because effluent is discharged to a subsurface recirculating gravel filter that is designed to resist plugging.

Because the wastewater treatment system flow rate is more than 20,000 gpd, Regional Water Board staff evaluated the need for a nitrogen effluent limit. Using the method contained in Attachment 1 of the General Order, Regional Water Board staff determined that a nitrogen effluent limit is not necessary because the combination of depth to groundwater and the reported percolation rate comply with the conditions presented in Table 5 of the General Order[1]. Although Regional Water Board staff do not propose an effluent limit for nitrogen in this case, the MRP requires that the College test for nitrate in discharged effluent in order to compare nitrate concentration in the effluent to nitrate concentrations in downgradient groundwater monitoring wells.

F. Groundwater Monitoring

This NOA and MRP R1-2018-0039 requires the College to monitor groundwater. Regional Water Board staff have determined that it is necessary to monitor groundwater downgradient of the effluent dispersal field, based on the proposed volume of discharge and historical effluent monitoring data for the Facility showing concentrations of arsenic, cadmium, hexavalent chromium, lead, thallium, carbon tetrachloride, tetrachloroethylene and vinyl chloride that exceeded Public Health Goals.

4. MONITORING AND REPORTING PROGRAM

The College shall comply with the attached MRP R1-2018-0039, for College of the Redwoods Wastewater Treatment and Disposal System, which is made part of this NOA by reference.

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[1] Attachment 1 of the General Order defines excessive percolation as "a combination of percolation rate and depth to ground water that does not comply with the conditions presented in Table 5." Bore hole testing performed at the disposal site resulted in a depth to groundwater of greater than 30 feet and the in-situ percolation test shows the percolation rate is 41.6 gpd/sf.

5. ENFORCEMENT

California Water Code section 13267 states, in part:

> "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports."

California Water Code section 13268 states, in part:

> "(a) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).
>
> (b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a violation of subdivision (a) in an amount which shall not exceed one thousand dollars ($1,000) for each day in which the violation occurs."

The reports required in item 3.B and 3.F are necessary to ensure that the College complies with the NOA and General Order. The burden, including costs of the reports, bears a reasonable relationship to the need for the reports and the benefits to be obtained from them. The reports required in this order are reasonable and are required by all dischargers enrolled under the General Order.

If Facility waste discharges violate the terms or conditions in the General Order, NOA and the MRP, the Regional Water Board may take enforcement action, including assessment of administrative civil liability. Discharge of wastes other than those described in the ROWD is prohibited. Prior to allowing changes to the wastewater strength or generation rate, or if the method of waste discharge changes from that described in the ROWD, the College must contact the Regional Water Board to determine if submittal of a new or revised ROWD is required.

The College will manage and discharge the waste subject to the terms and conditions of the General Order, NOA and the MRP and will maintain exclusive control over the discharges. As such, the College is responsible for compliance with the General Order and this NOA.

6. ANNUAL FEES & REGIONAL WATER BOARD CONTACT INFORMATION

The Facility's discharge is subject to payment of annual fees under the State Water Board's annual fee schedule for regulated discharges. Annual fees are based in part on ratings for threat to water quality and complexity. The subject Facility is rated 3-C and the College shall pay their annual fee for land disposal once the Facility begins discharging effluent.

The Fee Schedule is subject to change, and the fee for this Facility may change accordingly. Annual fees are due and payable as invoiced annually to the College by the State Water Board.

Regional Water Board Contact Information

The Regional Water Board's Groundwater Permitting Unit will manage regulation of the subject Facility under the General Order. Regional Water Board staff Justin McSmith, Water Resource Control Engineer, currently oversees this case. Please direct any questions or concerns about this NOA, the attached MRP, or related matters to Justin McSmith at (707) 576-2082 or Justin.McSmith@waterboards.ca.gov .

Sincerely,

Claudia Villacorta for

Digitally signed by Claudia Villacorta
Date: 2018.06.07 17:04:19 -07'00'

Water Boards

Matthias St. John
Executive Officer

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Enclosures:
Table 2, Summary of Applicable Discharge Requirements
Water Quality Order WQ 2014-0153-DWQ
Monitoring and Reporting Program R1-2018-0039

Attachments:
Attachment A, Site Location Map
Attachment B, Project Area Map
Attachment C, Wastewater Treatment System Schematic

cc: Steve McHaney, Steve.McHaney@ghd.com
Paul Cicerelli, Paul.Cicerelli@waterboards.ca.gov

TABLE 2, SUMMARY OF GENERAL ORDER REQUIREMENTS

General Order Section	Page	Numeric Limit	Requirement (refer to General Order and NOA for details)
A. PROHIBITIONS			
A.1-A.11	15	--	Prohibitions
B. REQUIREMENTS BY WASTEWATER SYSTEM TYPE			
B.1a-B.1.l	16	--	All Wastewater Systems
B.1.a	16	80,000 gpd	Wastewater Flow Limit (measured at headworks)
B.2.a-B.2.e	20	--	Septic Systems
B.6.a-B.6.h	24	--	Subsurface Disposal Systems
B.8.a-B.8.f	26	--	Sludge/Solids/Biosolids Disposal
D. EFFLUENT LIMITATIONS			
Wastewater Pond or Trickling Filter			

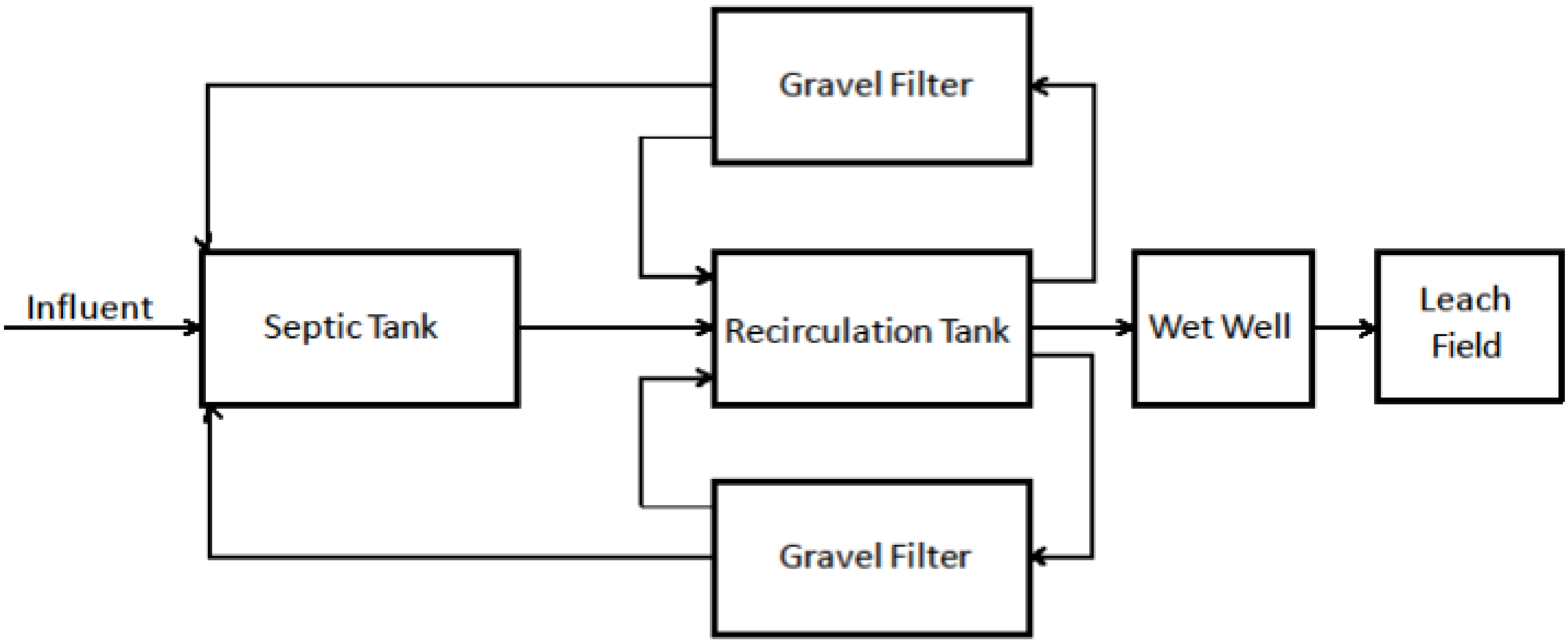
TABLE 2, SUMMARY OF GENERAL ORDER REQUIREMENTS

General Order Section	Page	Numeric Limit	Requirement (refer to General Order and NOA for details)
D.1.a	27	84 mg/L	Five Day Biochemical Oxygen Demand (BOD_5) (In accordance with Table 4 of the General Order and based on an influent concentration of 240 mg/L.)
D.1.a	27	Not applicable	Total Suspended Solids (TSS) (refer to General Order Table 4 for averaging allowances)

E. PROVISIONS			
Technical Report Preparation Requirements (list the reports below as appropriate)			
E.1.a	28	Have on-site.	Spill Prevention and Emergency Response Plan
E.1.b	29	Have on-site.	Sampling and Analysis Plan
E.1.c	30	Have on-site.	Sludge Management Plan
E.2.h	32	Have on-site.	A copy of NOA, General Order
E.2.k	32	Supervision	Certified wastewater treatment operator (when required)

ATTACHMENT A, SITE LOCATION MAP

ATTACHMENT B, PROJECT AREA MAP


Gravel Filter
Influent
Septic Tank
Recirculation Tank
Wet Well
Leach Field
Gravel Filter